Onion cells to an explanation

Year 7 · 7/SCI/01 · Tuesday 3 November 2026

Today's question: How does a cell's structure suit its job?

Success criteria: Pick a relevant feature > explain its effect > link that effect to the cell's job.

Work through the sections in order. Read the information before answering in your book, using the task IDs. Use the reading until the final exit check. Review in a different colour when instructed.

A1 · Start with yesterday's onion cells · 0-5 minutes

Yesterday you examined onion bulb skin using a microscope. These are plant cells, with a cell wall, cell membrane, cytoplasm and nucleus. The wall supports the cell. Onion bulb cells do not normally have chloroplasts. A drawing of your observation records only what you could actually see clearly.

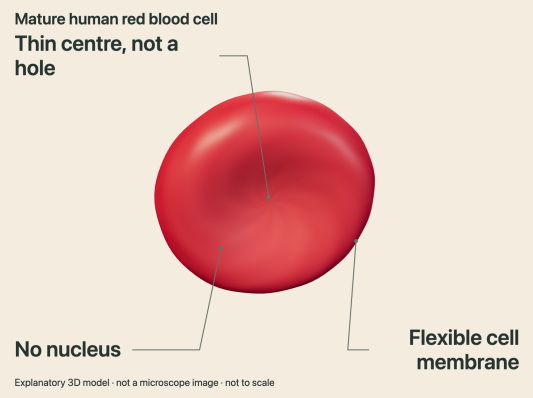
A1a. Name the structure that supports an onion cell. [1]

A1b. "These onion bulb cells have no chloroplasts, so they must be animal cells." Give one fact from the reading that shows why this conclusion is wrong. [1]

A2 · Learn the chain · 5-13 minutes

A specialised cell has features suited to a particular job. Cells do not consciously choose or decide. Explain cause and effect:

FEATURE > EFFECT (what changes) > JOB (how this helps)



A mature human red blood cell carries oxygen from the lungs to the body. It has no nucleus, leaving more room for haemoglobin, the substance that carries oxygen. It is a flexible disc, thin in the middle. Flexibility helps it squeeze through narrow blood vessels. Its thin centre gives oxygen a short distance to cross as it enters or leaves. The thin centre is not a hole. 3D teaching model, not to scale. Thin centre, not a hole.

Completed model: No nucleus > more room for haemoglobin > more oxygen can be carried.

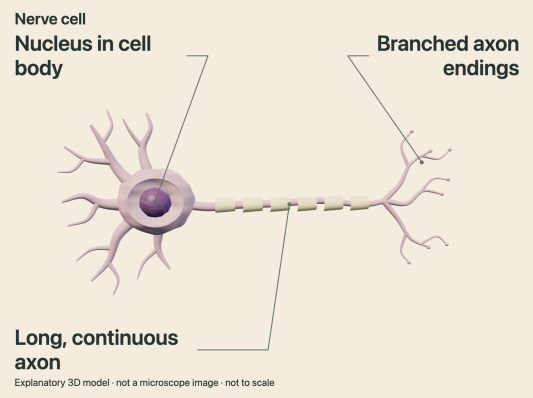
A2a · Unmarked. Copy the completed chain once. Label the feature, effect and job.

A2b · Unmarked. Correct: "Red blood cells decide to lose their nucleus because they want to carry oxygen." Include all three links.

Optional start: "A mature human red blood cell has ___, so ___, allowing it to ___."

Practise with a nerve cell

B1 · Reduce the help · 13-21 minutes



A nerve cell carries electrical signals. In this example, a signal travels from the spinal cord towards a muscle. The long axon allows a signal to travel a long distance along one cell. Branched endings form connections with several other cells, allowing the signal to be passed on at several contacts. The nerve cell has a nucleus. The axon and cell body are parts of the same cell. One nerve cell. 3D teaching model, not to scale.

B1a. Complete the missing effect: long axon > _____ > carries a signal between the spinal cord and a distant muscle. [1]

B1b. Write a complete feature > effect > job chain starting with branched endings. [3]

Optional help: What do the branches connect with? What can happen to the signal at those connections?

B2 · Check and repair · 21-26 minutes

B2a · Unmarked. Choose the best explanation. Give its letter.

A. A nerve cell has a long axon, so it is a different shape.

B. A nerve cell has a long axon, so a signal can travel a long distance along one cell to a distant muscle.

C. A nerve cell grows a long axon because it wants to reach a muscle.

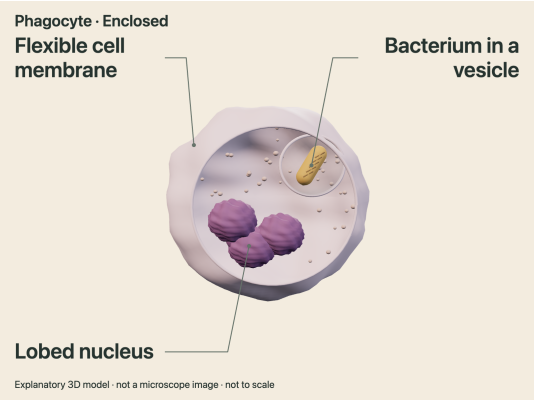
B2b · Unmarked. Explain what is missing or incorrect in one of the other statements.

B2c · Unmarked. When instructed, check A1-B2 against the model answers. Improve one answer in another colour. If all were correct, label the feature, effect and job in B1b.

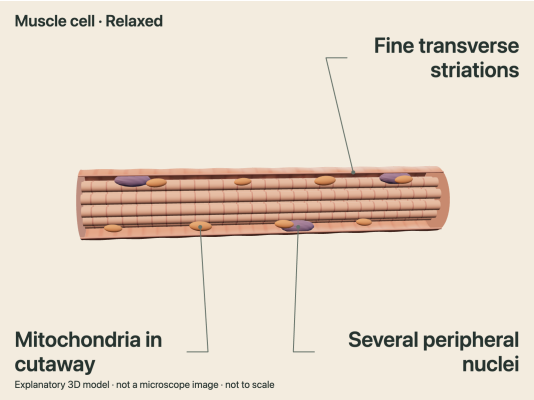
Your explanation checklist: Have I chosen the relevant feature? Have I explained what it changes? Have I linked that effect to the specific job? Have I avoided "wants", "tries" and "decides"?

Read, then explain two more cells

C · Read first · 26-31 minutes



White blood cell: White blood cells defend the body against pathogens, which can cause disease. A phagocyte is one type of white blood cell. Its flexible cell membrane changes shape, allowing it to flow around a bacterium and enclose it inside the cell. Chemicals inside the cell then digest and destroy the bacterium. Not all white blood cells work by engulfing bacteria. A phagocyte. 3D teaching model with an explanatory cutaway, not to scale.



Muscle cell: Muscle cells contract: they become shorter and pull, helping move part of the body. Contraction needs energy. Muscle cells contain many mitochondria, where aerobic respiration takes place. This process uses oxygen to release energy from food; it does not create energy. Many mitochondria help release energy at the rate needed for repeated contractions. Part of one skeletal muscle fibre. 3D teaching model with an explanatory cutaway, not to scale.

C · Independent explanations · begin at 31 minutes

These are original questions. Work alone. Use the information boxes; write the links in your own words.

C1. Explain how a flexible membrane helps a phagocyte defend the body against bacteria. [2]

C2. What do the chemicals do to the bacterium once it is enclosed? [1]

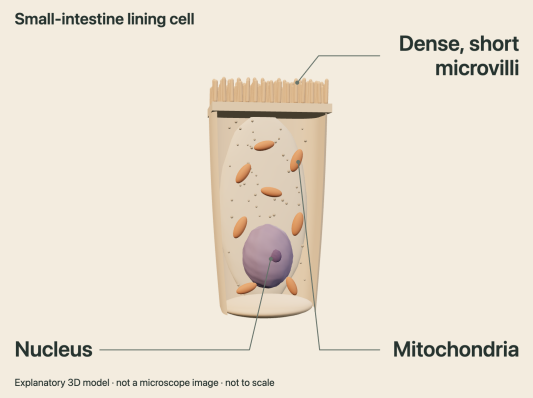
C3. Explain why muscle cells have many mitochondria. Include the process, what it provides, and what the muscle uses it for. [3]

Optional support: "The cell has ___, so ___, which helps it to ___." For C3 use respiration, releases energy, contract.

Continue to page 4. Complete C and D by approximately minute 42. Together these independent questions are worth 9 marks.

Apply, review and leave evidence

D · Fresh application · complete by minute 42



All the new facts you need: The small intestine absorbs digested food into the body. Each cell covering its inside surface has many tiny folds on its exposed surface. The folds give a larger surface area in contact with digested food. Surface area means the amount of outside surface. More membrane in contact with food provides more places where digested food can cross into the cell. This allows absorption at a higher rate. The nucleus and mitochondria inside the cell do not form its folds. One human intestinal lining cell. 3D teaching model with an explanatory cutaway, not to scale.

D1. Explain how this cell's surface structure helps it absorb digested food. Choose the relevant feature and write its full feature > effect > job chain. [3]

D2 · Review · 42-48 minutes · Unmarked. When instructed, check C1-D1. Improve a missing link or scientific word in another colour. If already correct, underline the effect in C1, C3 and D1. Leave a correct C2 unchanged.

F · Go further, if time · Unmarked

A proposed oxygen-carrying cell is long and branched, has a nucleus, has little room for haemoglobin and cannot bend easily.

F1. Give two linked limitations of this design for carrying oxygen through narrow blood vessels.

F2. Why is "the most complicated shape is always best" a poor rule? Use two cell examples from today.

E · Exit · final 2 minutes

Close or turn over the reading. Work independently. Leave these answers uncorrected for your teacher.

E1. Name the cell studied today that carries oxygen. [1]

E2. Complete: "In muscle cells, mitochondria are where _____ takes place." [1]

E3. Correct "All cells have a nucleus" using one precise example from today. [1]